Power

Power: USB-C, charger, regulator, LED switch

File: power.kicad_sch

MCU

MCU: Raytac MDBT50Q-1MV2 (nRF52840)

File: mcu.kicad_sch

Inputs

Inputs: 8 keys, 3 side buttons, 2 encoders

File: inputs.kicad_sch

Lighting

Lighting: level shifters, 8 key LEDs, ring strip

File: lighting.kicad_sch

Display

Display: 0.96 in 128x64 OLED (I2C)

File: display.kicad_sch

Mechanical

Mechanical: mounting, standoffs, fiducials

File: mechanical.kicad_sch

DESIGN NOTES (Rev A)
Power: USB-C -> PTC + USBLC6 -> BQ24074 (USB500: EN1=VBUS, EN2=GND) -> VSYS. TPS7A0533 (EN = power switch) -> +3V3. TPS22917 (ON = LED_EN) -> VLED.
MCU GPIO: keys 1-8 P0.14 P1.08 P0.28 P0.31 P0.12 P0.07 P0.27 P0.30 | L1 P1.01 L2 P0.21 SYS P0.29 | knob A P1.02/P0.25 sw P1.00 | knob B P1.15/P1.11 sw P1.13
I2C SDA P0.16 SCL P0.19 | LED ring P0.11, keys P0.05 (SPI MOSI; spare SCK pins are unconnected) | LED_EN P0.26 | BAT_SENSE P0.02 (AIN0) | CHG_N P0.23 PGOOD_N P1.03 | reset P0.18 | LFXO P0.00/P0.01
Unused, no-connect: P0.03 P0.04 P0.06 P0.08 P0.09 P0.10 P0.13 P0.15 P0.17 P0.20 P0.22 P0.24 P1.04 P1.05 P1.06 P1.07 P1.09 P1.10 P1.12 P1.14 (P0.09/P0.10 are NFC).
Key LED chain: keys 5-6-7-8-4-3-2-1; the back-row LEDs are turned 180 deg.

Checked against the TI BQ24074 datasheet: ITERM and TMR open = default 10% termination and default timers (R6, R7 not fitted);
TS 10k to VSS when the cell has no NTC; EN1 high + EN2 low = 450-500 mA input; K_ISET 890 A-ohm; OUT regulates at 4.4 V.

REVIEW BEFORE ORDERING
1 Battery J2 polarity must match the cell (JST-PH polarity is not standard).
2 OLED J5 pin order GND-VCC-SCL-SDA; header row position depends on the module drawing.
3 MDBT50Q-1MV2 (C5118826) was out at JLC on 23 Oct: use JLC Global Sourcing or consign it.
4 ST1-ST4 (Wurth 9774030243) are not stocked at LCSC: solder them by hand after assembly.
5 SK6812MINI-E needs VDD >= 3.7 V: firmware turns RGB off below about 3.7 V on battery.

Generated by tools/gen_schematic.py; edit the script
8-key wired + Bluetooth macro controller (see the plan page).

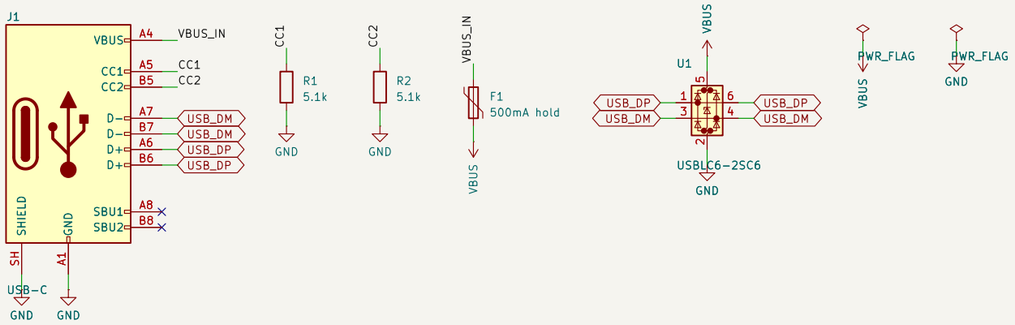
Deck-8

Sheet: /
File: deck8.kicad_sch

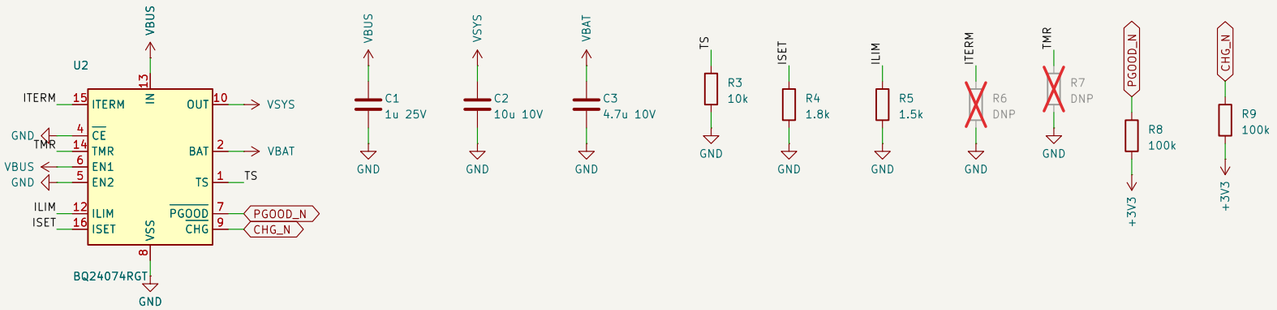
Title: Deck-8 Rev A — Overview

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/7

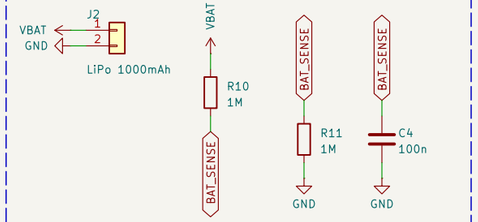
USB-C input and protection



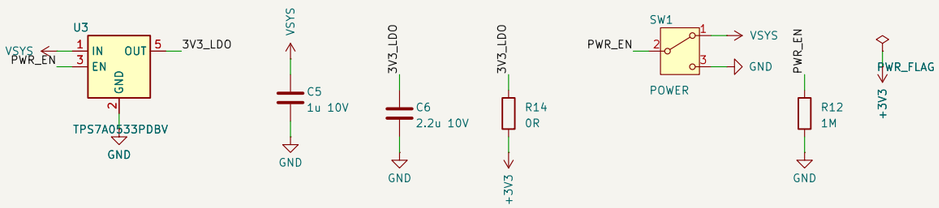
Charger (BQ24074, USB500 mode, 494 mA charge)



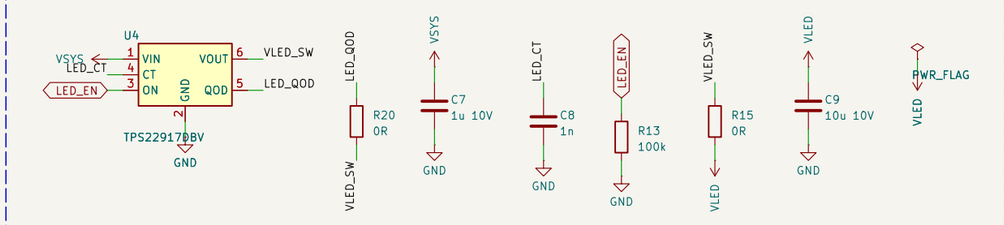
Battery and sense divider



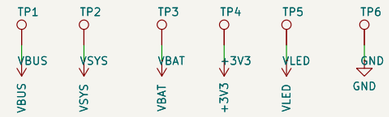
3.3 V regulator and power switch



LED power switch (all RGB off when dark)



Test points



Generated by tools/gen_schematic.py; edit the script

VSYS ~4.4 V on USB, else battery. Power switch = 3.3 V regulator enable.

Deck-8

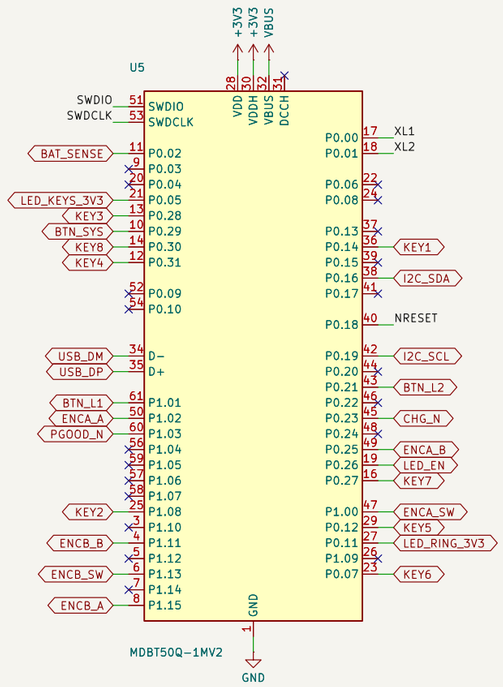
Sheet: /Power/

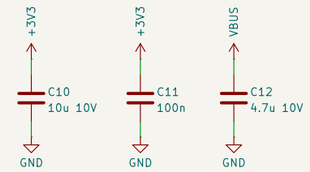
File: power.kicad_sch

Title: Deck-8 Rev A — Power: USB-C, charger, regulator, LED switch

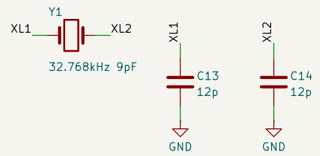
Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 2/7

nRF52840 module

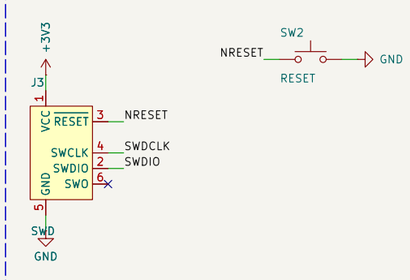




32.768 kHz crystal (LFXO)



Programming and reset



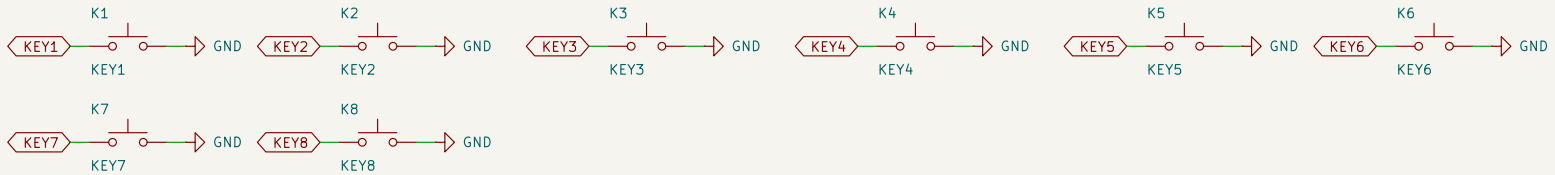
Generated by tools/gen_schematic.py; edit the script
Normal voltage mode: VDD + VDDH on +3V3, DCCH unused. GPIO plan: root sheet.
Deck−8

Sheet: /MCU/
File: mcu.kicad_sch

Title: Deck−8 Rev A — MCU: Raytac MDBT50Q−1MV2 (nRF52840)

Size: A3	Date: 2026−10−23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 3/7

Keys (MX hot-swap)



Side buttons



Encoders (knob A, knob B)



Generated by tools/gen_schematic.py; edit the script

Each input on its own GPIO, internal pull-up; any input wakes the MCU.

Deck-8

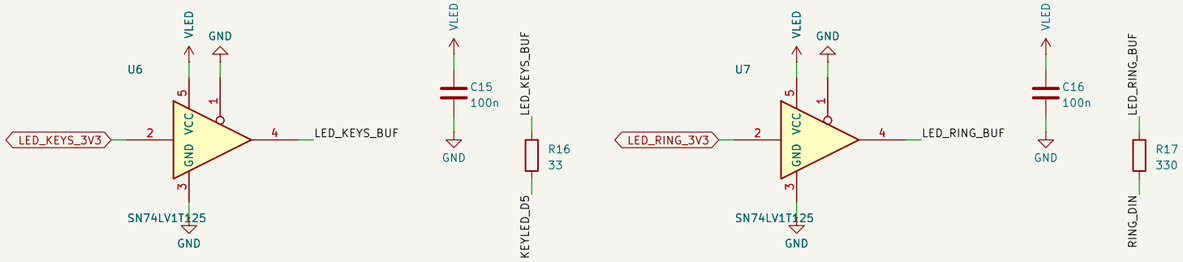
Sheet: /Inputs/

File: inputs.kicad_sch

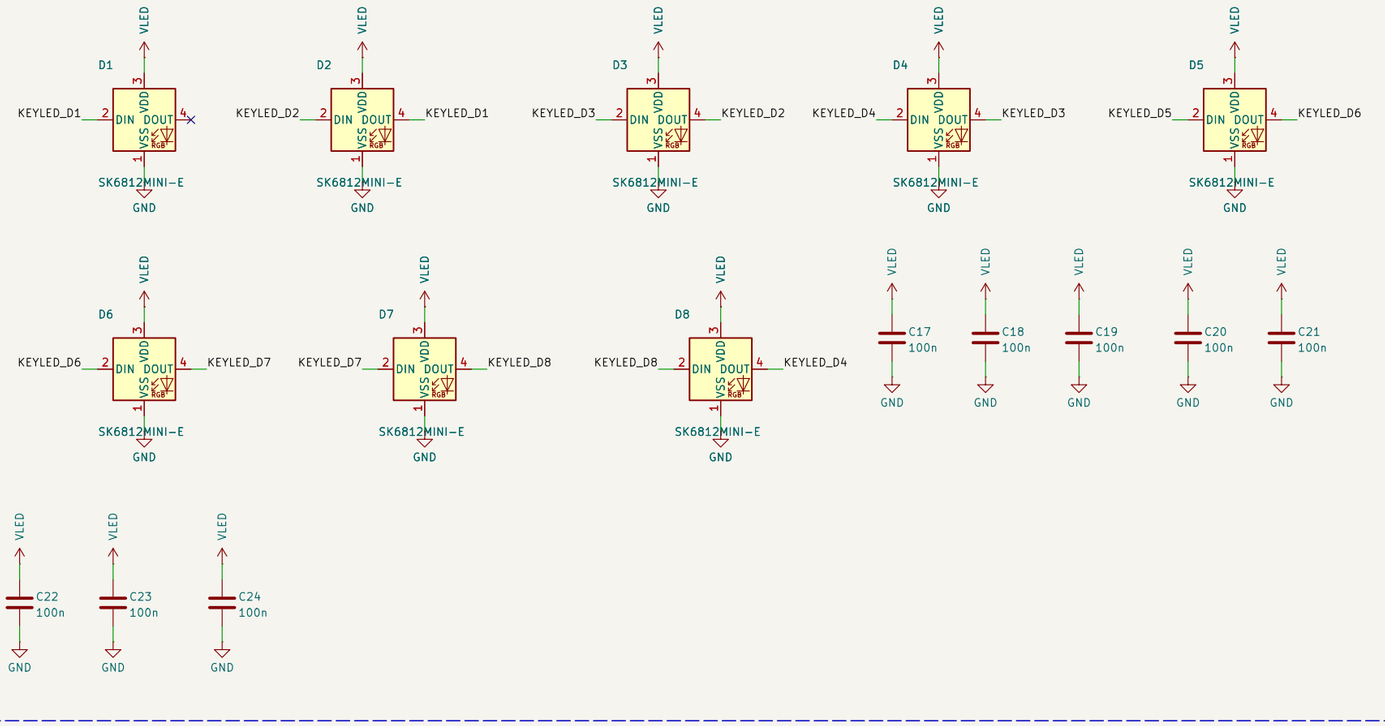
Title: Deck-8 Rev A — Inputs: 8 keys, 3 side buttons, 2 encoders

Size: A4	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 4/7

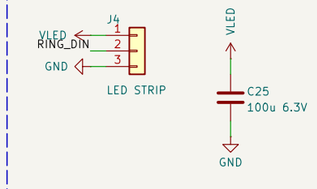
Level shifters (3.3 V data to VLED logic)



Key LEDs (chain: keys 5–6–7–8–4–3–2–1)



Ring strip connector



Generated by tools/gen_schematic.py; edit the script

Both chains on VLED (switched VSYS). Data held low while VLED is off.

Deck-8

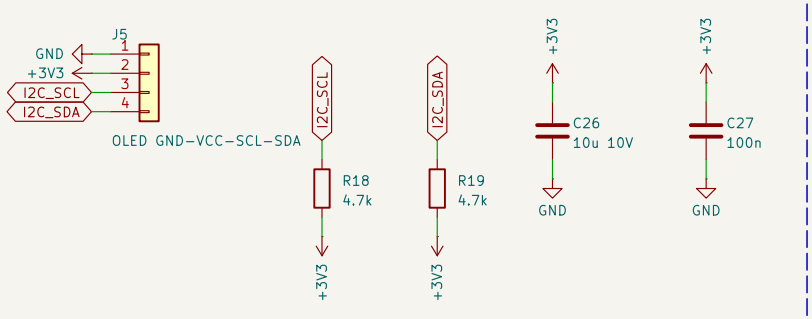
Sheet: /Lighting/

File: lighting.kicad_sch

Title: Deck-8 Rev A — Lighting: level shifters, 8 key LEDs, ring strip

Size: A3	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 5/7

OLED module header



Generated by tools/gen_schematic.py; edit the script

Header order GND, VCC, SCL, SDA. Buy modules with that order.

Deck-8

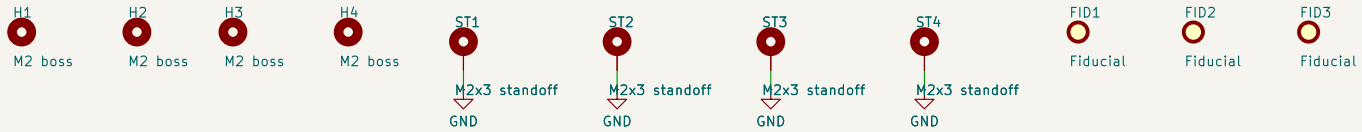
Sheet: /Display/

File: display.kicad_sch

Title: Deck-8 Rev A — Display: 0.96 in 128x64 OLED (I2C)

Size: A4	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 6/7

Mounting and fiducials



Generated by tools/gen_schematic.py; edit the script
Positions from the Fusion model (Deck8_Place field on each part).

Deck-8

Sheet: /Mechanical/
File: mechanical.kicad_sch

Title: Deck-8 Rev A — Mechanical: mounting, standoffs, fiducials

Size: A4	Date: 2026-10-23	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 7/7